APT3020 Knowledge-Based Systems

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Catalog Course Description

This course aims to enable students to design a knowledge structure integrated with production planning, quality control and other subsystems of an industrial organization. The course will also enable students to apply AI techniques to the problem of acquisition and representation of expert knowledge for problem solving in the expert's domain.

Credit: 4.5 Credits

Prerequisite: APT3010

Course Learning Outcomes

By the end of this course, students would be able to:

1. **Comprehend the foundational principles and diverse categories of Knowledge-Based Systems (KBS):** Students will be able to define KBS, identify their core components and characteristics, differentiate between various types such as expert systems, rule-based systems, and case-based systems, and recognize their applications in real-world scenarios.
2. **Master fundamental techniques for representing knowledge in AI systems:** Students will gain proficiency in utilizing logic-based approaches (propositional and first-order logic), semantic networks, frames, and ontologies to model and structure information for effective reasoning and problem-solving within KBS.
3. **Apply and evaluate different reasoning and inference mechanisms used in KBS:** Students will be able to understand and implement rule-based reasoning (forward and backward chaining), case-based reasoning, and basic methods for handling uncertainty, enabling them to derive new knowledge and make informed decisions using KBS.
4. **Analyze the architecture, development lifecycle, and tools associated with Expert Systems:** Students will develop a comprehensive understanding of expert systems as a significant type of KBS, including their components, knowledge acquisition methods, development stages, and the role of various tools and programming languages in their creation.
5. **Utilize and compare various problem-solving and search strategies within the context of KBS:** Students will learn about blind and heuristic search algorithms and their critical role in enabling KBS to navigate through potential solutions and efficiently address complex problems.
6. **Explore advanced and contemporary topics in KBS, including machine learning integration and semantic web technologies:** Students will gain an awareness of how machine learning techniques can enhance knowledge acquisition and refinement in KBS, and understand the principles and applications of semantic web technologies in building and utilizing large-scale knowledge systems.

Detailed Course Outline

#	Topic	Sub-Topics	Learning Outcomes	Capstone Assignment	References
1	**Introduction to Knowledge-Based Systems**	Definition of KBS, Characteristics of KBS, Components of KBS (Knowledge Base, Inference Engine), Types of KBS, Applications of KBS	1. **Remember:** Define the term "Knowledge-Based System". 2. **Understand:** Explain the core components of a KBS. 3. **Apply:** Identify different real-world applications of KBS. 4. **Analyze:** Differentiate between KBS and traditional computer programs. 5. **Evaluate:** Assess the benefits and limitations of using KBS. 6. **Create:** Propose a potential application of a KBS in a specific domain.	Write a brief report outlining the key characteristics and components of a Knowledge-Based System, providing an example of its application.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.smartsheet.com/knowledge-base-systems-and-templates
2	**Types of Knowledge-Based Systems**	Expert Systems, Rule-Based Systems, Case-Based Systems, Blackboard Systems, Intelligent Tutoring Systems, Classification Systems	1. **Remember:** List the different types of Knowledge-Based Systems. 2. **Understand:** Describe the fundamental principles of each type of KBS. 3. **Apply:** Classify a given scenario into one of the KBS types. 4. **Analyze:** Compare and contrast the strengths and weaknesses of different KBS types. 5. **Evaluate:** Determine the most suitable KBS type for a given problem. 6. **Create:** Design a high-level architecture for a specific type of KBS.	Develop a comparative table summarizing the key features, advantages, and disadvantages of at least three different types of Knowledge-Based Systems.	https://kmslh.com/glossary/knowledge-based-system/

3	**Knowledge Representation: Introduction**	Definition of Knowledge Representation, Importance of Knowledge Representation, Desirable Properties of Knowledge Representation, Different Approaches to Knowledge Representation	1. **Remember:** Define "Knowledge Representation" in the context of AI. 2. **Understand:** Explain why effective knowledge representation is crucial for KBS. 3. **Apply:** Identify the key properties that a good knowledge representation scheme should possess. 4. **Analyze:** Categorize different knowledge representation approaches based on their underlying principles. 5. **Evaluate:** Discuss the trade-offs involved in choosing a particular knowledge representation method. 6. **Create:** Outline a strategy for representing knowledge in a new, simple domain.	Prepare a presentation explaining the significance of knowledge representation and outlining the various approaches available.	https://www.geeksforgeeks.org/knowledge-representation-in-ai/, https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf, https://www.scaler.com/topics/artificial-intelligence-tutorial/techniques-of-knowledge-representation/
4	**Logic-Based Representation: Propositional Logic**	Syntax of Propositional Logic, Semantics of Propositional Logic, Logical Connectives, Truth Tables, Limitations of Propositional Logic	1. **Remember:** Define the basic symbols and connectives used in Propositional Logic. 2. **Understand:** Explain how truth values are assigned to propositions and compound statements. 3. **Apply:** Construct truth tables for given propositional logic expressions. 4. **Analyze:** Break down complex statements into propositional logic formulas. 5. **Evaluate:** Assess the expressiveness and limitations of Propositional Logic for knowledge representation. 6. **Create:** Formulate simple real-world scenarios using propositional logic.	Solve a set of logical puzzles by representing the given information in propositional logic and using truth tables to find the solutions.	https://www.geeksforgeeks.org/knowledge-representation-in-ai/

5	**Logic-Based Representation: First-Order Logic**	Syntax of First-Order Logic (Objects, Predicates, Variables, Quantifiers), Semantics of First-Order Logic, Universal and Existential Quantifiers, Representing Facts and Rules in FOL, Advantages of FOL over Propositional Logic	1. **Remember:** Identify the key components of First-Order Logic. 2. **Understand:** Explain the meaning and usage of quantifiers in FOL. 3. **Apply:** Translate simple natural language sentences into FOL. 4. **Analyze:** Compare and contrast Propositional Logic and First-Order Logic. 5. **Evaluate:** Discuss the increased expressiveness of FOL for knowledge representation. 6. **Create:** Represent a more complex scenario using first-order logic.	Translate a given set of natural language rules and facts about a family relationship into First-Order Logic.	https://www.geeksforgeeks.org/knowledge-representation-in-ai/
6	**Inference in First-Order Logic**	Inference Rules (Modus Ponens, Universal Instantiation, etc.), Forward Chaining, Backward Chaining, Resolution, Unification	1. **Remember:** List common inference rules used in First-Order Logic. 2. **Understand:** Explain the mechanisms of forward and backward chaining. 3. **Apply:** Use inference rules to derive new conclusions from a set of FOL statements. 4. **Analyze:** Determine the steps involved in the resolution inference process. 5. **Evaluate:** Compare the efficiency and applicability of forward and backward chaining. 6. **Create:** Design a simple inference process for a given set of FOL rules.	Given a knowledge base in FOL, apply forward or backward chaining to answer specific queries.	https://www.geeksforgeeks.org/knowledge-representation-in-ai/

7	**Semantic Networks**	Structure of Semantic Networks (Nodes, Edges, Labels), Types of Relationships (Is-A, Part-Of, etc.), Inheritance in Semantic Networks, Advantages and Disadvantages of Semantic Networks	1. **Remember:** Describe the basic components of a Semantic Network. 2. **Understand:** Explain how different types of relationships are represented in semantic networks. 3. **Apply:** Draw a semantic network to represent a given set of facts. 4. **Analyze:** Identify the hierarchical and associative relationships within a semantic network. 5. **Evaluate:** Discuss the strengths and weaknesses of using semantic networks for knowledge representation. 6. **Create:** Design a semantic network for a specific domain, identifying key concepts and their relationships.	Model a simple biological taxonomy using a semantic network diagram.	http://www.eecs.qmul.ac.uk/~mmh/AINotes/AINotes4.pdf,(https://web.stanford.edu/class/cs227/Readings/KryptonAFunctionalApproachToKRR.pdf), https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf, https://www.geeksforgeeks.org/semantic-networks-in-artificial-intelligence/, https://www.scaler.com/topics/artificial-intelligence-tutorial/techniques-of-knowledge-representation/, https://www.professional-ai.com/difference-between-semantic-net-and-frame.html

8	**Frames**	Structure of Frames (Slots, Facets, Fillers), Inheritance in Frame Systems, Default Values, Procedures Attached to Frames, Advantages and Disadvantages of Frames	1. **Remember:** Identify the main components of a frame in knowledge representation. 2. **Understand:** Explain how inheritance works in frame-based systems. 3. **Apply:** Represent a stereotypical object or event using a frame structure. 4. **Analyze:** Compare and contrast frames with semantic networks. 5. **Evaluate:** Discuss the suitability of frames for representing specific types of knowledge. 6. **Create:** Design a set of interconnected frames to represent a common scenario.	Develop a frame-based representation for a "Student" entity, including relevant slots, facets, and default values.	http://www.eecs.qmul.ac.uk/~mmh/AINotes/AINotes4.pdf,(https://web.stanford.edu/class/cs227/Readings/KryptonAFunctionalApproachToKRR.pdf), https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf, https://www.geeksforgeeks.org/frames-in-ai-knowledge-representation-and-inheritance/, https://www.scaler.com/topics/artificial-intelligence-tutorial/techniques-of-knowledge-representation/, https://www.professional-ai.com/difference-between-semantic-net-and-frame.html
9	**Ontologies: Definition and Purpose**	Definition of Ontology in KBS, Purpose of Ontologies (Shared Understanding, Knowledge Sharing, Reasoning), Components of an Ontology (Classes, Properties, Relationships,	1. **Remember:** Define the concept of an ontology in the context of Knowledge-Based Systems. 2. **Understand:** Explain the key purposes and benefits of using ontologies. 3. **Apply:** Identify the core components that typically constitute an ontology. 4. **Analyze:** Differentiate between different levels or types of ontologies. 5. **Evaluate:** Discuss the role of ontologies in facilitating knowledge sharing and reasoning.	Write a short essay explaining the definition and purpose of ontologies in knowledge representation.	https://www.smartsheet.com/knowledge-base-systems-and-templates,(https://www.researchgate.net/publication/307863010_Knowledge_Representation_with_Ontology_Tools_Methodology), https://ischool.illinois.edu/research/areas/knowledge-representation-ontologies, https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=c0cf4b0a15db9e60811900d4a00f4

		Axioms), Levels of Ontologies	6. **Create:** Outline the basic structure of an ontology for a given domain.		481a251b35b, https://www.abdn.ac.uk/registry/courses/undergraduate/2019-2020/computing_science/cs3025, https://www4.uib.no/en/courses/info282, https://telnyx.com/learn-ai/knowledge-reasoning
10	**Ontology Development**	Ontology Development Methodologies (e.g., Top-Down, Bottom-Up, Middle-Out), Key Steps in Ontology Development (Conceptualization, Formalization, Implementation, Evaluation), Ontology Languages (RDF, RDFS, OWL)	1. **Remember:** List the common methodologies for developing ontologies. 2. **Understand:** Describe the main steps involved in the ontology development process. 3. **Apply:** Identify appropriate ontology languages for different knowledge representation needs. 4. **Analyze:** Compare and contrast different ontology development methodologies. 5. **Evaluate:** Assess the challenges and best practices in building effective ontologies. 6. **Create:** Develop a conceptual model for a simple ontology in a chosen domain.	Design the initial conceptualization phase of an ontology for a specific domain (e.g., university courses).	https://en.wikipedia.org/wiki/Knowledge-based_systems,(https://www.researchgate.net/publication/307863010_Knowledge_Representation_with_Ontology_Tools_Methodology)

11	**Semantic Web Technologies and KBS**	Introduction to the Semantic Web, Role of Ontologies in the Semantic Web, RDF (Resource Description Framework), RDFS (RDF Schema), OWL (Web Ontology Language), SPARQL Query Language	1. **Remember:** Identify the core technologies associated with the Semantic Web. 2. **Understand:** Explain how ontologies are utilized within the Semantic Web framework. 3. **Apply:** Describe the basic principles of RDF, RDFS, and OWL. 4. **Analyze:** Explain how SPARQL is used to query semantic data. 5. **Evaluate:** Discuss the impact of Semantic Web technologies on the development of KBS. 6. **Create:** Formulate a simple query using SPARQL for a given RDF dataset.	Write a report on how Semantic Web technologies enhance the capabilities of Knowledge-Based Systems.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.geeksforgeeks.org/knowledge-representation-in-ai/
12	**Reasoning and Inference in KBS**	Rule-Based Reasoning (Forward Chaining, Backward Chaining), Case-Based Reasoning, Reasoning with Uncertainty (Introduction to Bayesian Reasoning, Fuzzy Logic)	1. **Remember:** List different types of reasoning mechanisms used in KBS. 2. **Understand:** Explain the step-by-step process of forward and backward chaining. 3. **Apply:** Determine whether forward or backward chaining is more suitable for a given problem. 4. **Analyze:** Compare and contrast rule-based and case-based reasoning. 5. **Evaluate:** Discuss the challenges of reasoning with uncertain information. 6. **Create:** Design a simple rule-based system for a specific task.	Given a set of rules and facts, trace the execution of forward or backward chaining to reach a conclusion.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.geeksforgeeks.org/knowledge-representation-in-ai/,(https://web.mst.edu/lib-circ/files/Special%20Collections/INCOSE2010/An%20introduction%20to%20knowledge%20representation%20and%20ontology%20development%20for%20systems%20engineers.pdf)

13	**Expert Systems: Architecture and Components**	Basic Architecture of an Expert System (Knowledge Base, Inference Engine, User Interface), Knowledge Acquisition Subsystem, Explanation Subsystem, Working Memory	1. **Remember:** Identify the main components of an Expert System. 2. **Understand:** Explain the function of each component in the overall architecture. 3. **Apply:** Describe how knowledge is acquired and utilized in an expert system. 4. **Analyze:** Explain the role of the explanation subsystem in enhancing user trust. 5. **Evaluate:** Discuss the key architectural considerations in designing an expert system. 6. **Create:** Sketch a basic architecture for an expert system in a chosen domain.	Draw a diagram illustrating the architecture of a typical expert system and briefly describe the role of each component.	https://kmslh.com/glossary/knowledge-based-system/, https://en.wikipedia.org/wiki/Knowledge-based_systems,(https://web.mst.edu/lib-circ/files/Special%20Collections/INCOSE2010/An%20introduction%20to%20knowledge%20representation%20and%20ontology%20development%20for%20systems%20engineers.pdf), https://www.slideshare.net/slideshow/introduction-to-ontologies-knowledge-representation-reasoning/273144559
14	**Expert Systems: Knowledge Acquisition**	Challenges in Knowledge Acquisition, Techniques for Knowledge Acquisition (Interviews, Protocol Analysis, Questionnaires, Rule Induction), Role of the Knowledge Engineer	1. **Remember:** List common challenges encountered during knowledge acquisition. 2. **Understand:** Describe various techniques used to elicit knowledge from domain experts. 3. **Apply:** Identify the appropriate knowledge acquisition technique for a given scenario. 4. **Analyze:** Explain the crucial role of the knowledge engineer in the development process. 5. **Evaluate:** Discuss the importance of effective communication between the	Conduct a mock interview with a classmate acting as a domain expert and document the acquired knowledge.	https://en.wikipedia.org/wiki/Knowledge-based_systems,(https://web.mst.edu/lib-circ/files/Special%20Collections/INCOSE2010/An%20introduction%20to%20knowledge%20representation%20and%20ontology%20development%20for%20systems%20engineers.pdf)

			knowledge engineer and the domain expert. 6. **Create:** Develop a set of interview questions to acquire knowledge from a hypothetical domain expert.		
15	**Expert Systems: Development Lifecycle and Tools**	Stages in the Development Lifecycle of an Expert System (Problem Identification, Knowledge Acquisition, Design, Implementation, Testing, Deployment, Maintenance), Introduction to Expert System Shells (e.g., CLIPS, Exsys), Programming Languages for Expert Systems (e.g., Prolog)	1. **Remember:** List the key stages in the expert system development lifecycle. 2. **Understand:** Explain the activities involved in each stage of the lifecycle. 3. **Apply:** Identify suitable tools and programming languages for building expert systems. 4. **Analyze:** Discuss the iterative nature of the expert system development process. 5. **Evaluate:** Assess the importance of testing and maintenance in ensuring the reliability of an expert system. 6. **Create:** Outline a development plan for a simple expert system.	Research and present on a specific expert system shell or programming language used for building expert systems.	https://en.wikipedia.org/wiki/Knowledge-based_systems,(https://web.mst.edu/lib-circ/files/Special%20Collections/INCOSE2010/An%20introduction%20to%20knowledge%20representation%20and%20ontology%20development%20for%20systems%20engineers.pdf), https://www.geeksforgeeks.org/knowledge-representation-in-ai/, https://www.slideshare.net/slideshow/introduction-to-ontologies-knowledge-representation-reasoning/273144559,(https://www.researchgate.net/publication/307863010_Knowledge_Representation_with_Ontology_Tools_Methodology)

16	**Problem Solving and Search in KBS**	Introduction to Problem Solving in AI, State Space Representation, Blind Search Strategies (Breadth-First Search, Depth-First Search), Heuristic Search Strategies (Best-First Search, A* Search), Role of Search in KBS	1. **Remember:** Define the concept of a state space in problem solving. 2. **Understand:** Explain the principles of blind and heuristic search algorithms. 3. **Apply:** Trace the execution of a simple search algorithm on a given problem. 4. **Analyze:** Compare and contrast different search strategies in terms of efficiency and completeness. 5. **Evaluate:** Discuss the importance of heuristics in solving complex problems in KBS. 6. **Create:** Formulate a search problem and propose a suitable search strategy.	Implement a simple search algorithm (e.g., BFS or DFS) to solve a basic problem like navigating a grid.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.geeksforgeeks.org/knowledge-representation-in-ai/, https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf, https://www.slideshare.net/slideshow/introduction-to-ontologies-knowledge-representation-reasoning/273144559
17	**Machine Learning for Knowledge Acquisition**	Introduction to Machine Learning, Symbolic Machine Learning Methods (Decision Trees, Rule Learning), Analogy-Based Learning (Case-Based Reasoning), Clustering for Knowledge Discovery, Incremental Concept Learning	1. **Remember:** Define the basic concepts of machine learning. 2. **Understand:** Explain how symbolic machine learning techniques can be used for knowledge acquisition. 3. **Apply:** Identify scenarios where different machine learning methods can be used to extract knowledge. 4. **Analyze:** Compare and contrast different symbolic machine learning approaches. 5. **Evaluate:** Discuss the advantages and limitations of using machine learning for knowledge acquisition in KBS. 6. **Create:** Outline a plan to use a specific machine learning technique to acquire knowledge in a given domain.	Explore a publicly available dataset and propose how a machine learning algorithm could be used to extract relevant knowledge for a KBS.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.geeksforgeeks.org/knowledge-representation-in-ai/, https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf, https://www.slideshare.net/slideshow/introduction-to-ontologies-knowledge-representation-reasoning/273144559

18	**Case-Based Reasoning in Detail**	The CBR Cycle (Retrieve, Reuse, Revise, Retain), Case Representation, Similarity Measures, Case Adaptation, Advantages and Disadvantages of CBR	1. **Remember:** List the steps involved in the Case-Based Reasoning cycle. 2. **Understand:** Explain how cases are represented and retrieved in a CBR system. 3. **Apply:** Identify appropriate similarity measures for different types of cases. 4. **Analyze:** Describe the process of adapting a retrieved case to a new problem. 5. **Evaluate:** Discuss the strengths and weaknesses of using CBR for problem solving. 6. **Create:** Design a basic case representation for a specific problem domain.	Develop a simple case base for a specific problem (e.g., recommending movies) and outline the steps to solve a new problem using CBR.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.geeksforgeeks.org/knowledge-representation-in-ai/, https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf
19	**Planning in Knowledge-Based Systems**	Introduction to Planning, Components of a Planning System (Goal, Actions, States), Forward and Backward Planning, Plan Representation, Applications of Planning in KBS	1. **Remember:** Define the concept of planning in the context of AI. 2. **Understand:** Explain the key components of a planning system. 3. **Apply:** Distinguish between forward and backward planning approaches. 4. **Analyze:** Describe how plans are typically represented. 5. **Evaluate:** Discuss the challenges and applications of planning in KBS. 6. **Create:** Formulate a simple planning problem and propose a sequence of actions to achieve the goal.	Model a simple task (e.g., making a cup of tea) as a planning problem, defining the initial state, goal state, and possible actions.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.geeksforgeeks.org/knowledge-representation-in-ai/, https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf, https://www.slideshare.net/slideshow/introduction-to-ontologies-knowledge-representation-reasoning/273144559

20	**Constraint Satisfaction Problems**	Definition of CSP, Components of a CSP (Variables, Domains, Constraints), Constraint Propagation, Backtracking Search, Applications of CSP in KBS	1. **Remember:** Define the key elements of a Constraint Satisfaction Problem. 2. **Understand:** Explain the concepts of constraint propagation and backtracking search. 3. **Apply:** Model a simple real-world problem as a CSP. 4. **Analyze:** Describe how constraints are used to reduce the search space in CSPs. 5. **Evaluate:** Discuss the suitability of CSPs for solving certain types of problems in KBS. 6. **Create:** Formulate a set of constraints for a given problem.	Model a simple scheduling problem as a Constraint Satisfaction Problem, identifying variables, domains, and constraints.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.geeksforgeeks.org/knowledge-representation-in-ai/, https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf, https://www.slideshare.net/slideshow/introduction-to-ontologies-knowledge-representation-reasoning/273144559
21	**Knowledge Revision and Truth Maintenance**	Challenges of Maintaining Consistency in Knowledge Bases, Sources of Knowledge Change, Truth Maintenance Systems (TMS), Belief Revision, Non-Monotonic Reasoning	1. **Remember:** Identify the challenges associated with updating knowledge bases. 2. **Understand:** Explain the purpose of Truth Maintenance Systems. 3. **Apply:** Describe how beliefs can be revised in response to new information. 4. **Analyze:** Differentiate between monotonic and non-monotonic reasoning. 5. **Evaluate:** Discuss the importance of knowledge revision in dynamic environments. 6. **Create:** Outline a simple scenario where knowledge revision would be necessary.	Describe a situation where new information contradicts existing knowledge in a KBS and explain how a truth maintenance system might handle this conflict.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf

22	**Intelligent Agents and KBS**	Definition of Intelligent Agents, Characteristics of Intelligent Agents (Autonomy, Reactivity, Proactiveness, Social Ability), Integration of KBS with Intelligent Agents, Agent Architectures	1. **Remember:** Define the concept of an Intelligent Agent. 2. **Understand:** Explain the key characteristics that define an intelligent agent. 3. **Apply:** Describe how knowledge-based systems can be used to build intelligent agents. 4. **Analyze:** Compare different architectures for intelligent agents. 5. **Evaluate:** Discuss the benefits of integrating KBS with intelligent agents. 6. **Create:** Propose an architecture for an intelligent agent that utilizes a knowledge-based system.	Design a simple intelligent agent for a specific task, outlining how it would use a knowledge base to achieve its goals.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.csee.umbc.edu/courses/undergraduate/471/fall15/02/static/files/krr.pdf
23	**Evaluating and Validating Knowledge-Based Systems**	Importance of Evaluation and Validation, Criteria for Evaluation (Accuracy, Efficiency, Usability, Completeness), Methods for Evaluation (Testing, Expert Review, User Feedback), Validation Techniques	1. **Remember:** Explain why evaluating and validating KBS are important. 2. **Understand:** Describe the key criteria used to assess the quality of a KBS. 3. **Apply:** Identify different methods for evaluating the performance of a KBS. 4. **Analyze:** Explain the difference between validation and verification. 5. **Evaluate:** Discuss the challenges in objectively evaluating the effectiveness of a KBS. 6. **Create:** Develop a basic evaluation plan for a hypothetical KBS.	Design a set of test cases to evaluate the performance of a rule-based expert system for a specific domain.	https://en.wikipedia.org/wiki/Knowledge-based_systems,(https://web.mst.edu/lib-circ/files/Special%20Collections/INCOSE2010/An%20introduction%20to%20knowledge%20representation%20and%20ontology%20development%20for%20systems%20engineers.pdf)

24	**Emerging Trends and Future Directions in KBS**	Integration of KBS with Machine Learning (Deep Learning, Knowledge Graphs), Scalable Knowledge Graphs, Explainable AI (XAI) in KBS, Human-AI Collaboration with KBS, Applications in Emerging Domains	1. **Remember:** Identify current trends in the field of Knowledge-Based Systems. 2. **Understand:** Explain how KBS are being integrated with other AI technologies like machine learning. 3. **Apply:** Describe the concept of Knowledge Graphs and their role in KBS. 4. **Analyze:** Discuss the importance of explainability in modern KBS. 5. **Evaluate:** Assess the potential of KBS in future applications and research. 6. **Create:** Propose a novel application of KBS leveraging emerging trends in AI.	Research and present on a specific emerging trend in Knowledge-Based Systems, such as the use of knowledge graphs or explainable AI.	https://en.wikipedia.org/wiki/Knowledge-based_systems, https://www.geeksforgeeks.org/knowledge-representation-in-ai/

Teaching Methodology

This course employs a **blended, student-centered approach** that prioritizes **active learning** and **hands-on practice**. Key strategies include:

1. **Scaffolded Learning**: Concepts are introduced incrementally, starting with foundational syntax and progressing to advanced applications. Each lesson builds on prior knowledge through guided examples and exercises.
2. **Interactive Coding Labs**: Students engage in real-time coding exercises during class, with immediate feedback from instructors to reinforce understanding and troubleshoot errors.
3. **Collaborative Learning**: Pair programming, group projects, and peer code reviews foster teamwork and expose students to diverse problem-solving approaches.
4. **Project-Based Learning**: Capstone assignments and a final project simulate real-world challenges, requiring students to synthesize concepts, debug code, and present solutions.
5. **Flipped Classroom Elements**: Pre-class resources (e.g., short videos, readings) introduce theoretical concepts, freeing class time for interactive problem-solving and discussion.

Assessment & Feedback:

- **Formative assessments** (quizzes, mini-projects) track progress.
- **Summative assessments** (capstone assignments, final project) evaluate mastery.
- **Differentiated instruction** accommodates varying skill levels through optional challenges and tailored support.

This methodology ensures students develop **technical proficiency**, **critical thinking**, and **confidence** to tackle real-world programming tasks.

Assessment

Mid Semester Exam	20%
End Semester Exam	30%
Portfolio	15%
Group work	10%
CAT	5%
Project	15
Attendance/Particiapation	5%

Grading System

Score	Grade
90 - 100	A
87 -89	A-
84 - 86	B+
80 - 83	B
77 - 79	B-
74 - 76	C+
70 - 74	C
67 - 69	C-
64 - 66	D+
62 - 63	D
60 - 61	D-
0 - 59	F

Accessibility & Inclusion Statement

USIU-Africa is committed to ensuring that all students, including those with disabilities, have equitable access to learning opportunities. If you anticipate or experience any barriers to learning, please contact the Disability Inclusion Department at inclusion@usiu.ac.ke. You are also encouraged to speak with me so we can work together to ensure you have the support you need to succeed in this course.